

Yuchi Liu

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EDUCATION

Emory University

Atlanta, USA

Expected MSPH. in **Biostatistics**

Aug. 2024 – Present

- **Core Courses:** SAS Programming, R Programming, Machine Learning, Data Science Toolkit, Biostatistical Methods, Statistical inference, Modern Regression Analysis, Clinical Trials, Biostatistical Consulting, Data Science (SQL)

The University of Liverpool

Liverpool, U.K.

B.S. in **Mathematics**

Sep. 2022 – Jun. 2024

- **Core Courses:** Statistics and Probability I&II, Vector Calculus, Operational Research, Financial Mathematics

Xi'an Jiaotong - Liverpool University (XJTLU)

Suzhou, China

B.S. in **Applied Mathematics**

Sep. 2020 – Jun. 2022

- **Core Courses:** Calculus (Mathematical Science), Multivariable Calculus, Analysis 1&2, Dynamic Modelling

RESEARCH EXPERIENCE

Research on the Validity of Regional Group IQ Data | Student Researcher

Oct. 2022 - Jun. 2023

- Oversaw the verification of data validity for a project on the impact of regional IQ levels on regional development, validated a dataset consisting of 800,000 records and conducted the Kolmogorov-Smirnov test using SPSS to assess data normality
- Used SPSS for exploratory data analysis, visualized trends and distributions of fluid intelligence and crystallized intelligence, and employed regression analysis models to examine the relationship between the distribution disparity of intelligence and local economic development, and supported the report writing

AI-Based Medical Triage System for Oculoplastics Referral | Research Assistant

Feb. 2026 – Present

- Lead the statistical analysis for a first-of-its-kind fine-tuned LLM designed to assist in medical referral triage
- Use Weighted Kappa and AUROC to quantify AI performance, validating that the model achieved "near expert level"
- Using R (confusion matrix, error distribution, and ROC curves) to visualize complex datasets, the safety and diagnostic accuracy of artificial intelligence are clearly demonstrated to clinical stakeholders
- Demonstrate model generalizability and safety by maintaining 100% sensitivity in identifying high-priority emergencies

Global Health Data Analysis (COVID-19 Focus)

Sep. 2025-Dec. 2025

- Designed a 4-table relational database (ERD) to manage and normalize a global health dataset containing 1,000,000 rows
- Developed queries using subqueries and left joins to analyze mortality and recovery trends across multiple countries.
- Quantified treatment success rates and identified that the 36–60 age group experienced the highest COVID-19 burden in 2020, while Hepatitis led all tracked diseases in mortality rate at 5.22%.

Sleep Health and Lifestyle Analysis

Sep. 2024 - Dec. 2024

- Analyzed the Sleep Health Dataset from Kaggle to examine factors affecting sleep quality using statistical modeling in R
- Investigated the impact of age, gender, BMI, blood pressure, and heart rate on sleep quality using ANOVA and linear regression and assessed the relationship between physical activity and sleep health via bivariate regression
- Performed Kruskal-Wallis and chi-square tests to determine group differences and built a logistic regression model to quantify the association between stress levels and sleep disorder presence
- Identified key factors affecting sleep quality, with significant associations found for BMI, blood pressure, and stress levels

INTERNSHIP EXPERIENCE

Yangzhou Rural Commercial Bank | Data Analysis Intern

Jun. 2022 - Aug. 2022

- Responsible for data governance in a data tagging management platform and provided data support for the front office
- Analyzed customer identification requirements, used logic tree analysis and user behavior analysis models to conduct customer segmentation analysis, and generated customer lists for business departments to develop promotional plans
- Collected feedback from business departments to further optimize and adjust the tagging data, and aided business departments in improving their overall sales performance, especially in closed-end funds and T+1-based products
- Cleaned data and performed departmental report data analysis using Python to ensure standardized usage of the platform

COMPETITIONS

Kaggle Competition: CommonLit - Evaluate Student Summaries | Group leader

Jul. 2023 - Oct. 2023

- Conducted Exploratory Data Visualization to identify the relations between the features and content and wording scores of summaries of passages written by students in grades 3-12 on a variety of topics and genres
- Developed feature engineering to add a lot of new features to boost the power of machine-learning models
- Implemented and evaluated different machine learning models like DeBERTa, XGBoost, and CatBoost, used Optuna to optimize the hyperparameters of three tree models based on RMSE of the validation dataset, performed 4-group fold cross-validation to reduce overfitting, and implemented ensemble strategy to combine the best models for the final prediction
- Achieved top 8% in the public leaderboard

ADDITIONAL INFORMATION

Computer Skills: SQL, JAVA, Python, R, SPSS, SAS

Language: English, Mandarin